

ISTANBUL SHOPPING ANALYTICS

Business Intelligence Report

A Data-Driven Analysis of Customer Behavior, Sales Performance
& Revenue Trends Across 10 Istanbul Shopping Malls

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Date: July 2026

Dataset Period: January 2021 – March 2023

Total Transactions Analyzed	99,457
Total Revenue Tracked	\$251,505,794
Shopping Malls Covered	10 Malls
Data Period	Jan 2021 – Mar 2023

1. Executive Summary

This report presents a comprehensive analysis of shopping behavior and revenue performance across 10 major shopping malls in Istanbul, Turkey, covering the period from January 2021 to March 2023. The analysis is based on 99,457 customer transactions totaling \$251,505,794 in revenue.

The findings are intended to support shopping mall management in making data-driven decisions around tenant mix, marketing strategy, customer targeting, and operational planning.

Key Finding

Clothing dominates all revenue categories across every mall, female shoppers drive nearly 60% of total revenue, and Seniors aged 56+ represent the highest-spending customer segment despite being a smaller demographic group.

Business Questions Investigated

- **Q1.** Which product categories generate the most revenue across Istanbul malls?
- **Q2.** How does customer gender influence purchasing behavior and revenue contribution?
- **Q3.** Which age groups represent the most valuable customer segments?
- **Q4.** What payment methods do customers prefer and what does this reveal about shopping habits?
- **Q5.** Which shopping malls are the top revenue performers and why?
- **Q6.** How has revenue trended over time and what seasonal patterns exist?
- **Q7.** Are there anomalies or risks in the data that management should be aware of?

2. Dataset Overview & Methodology

The dataset used for this analysis was sourced from Kaggle (Customer Shopping Dataset — Istanbul) and contains transactional records from 10 shopping malls across Istanbul. Prior to analysis, the dataset underwent thorough cleaning and preparation.

Data Cleaning Steps

- Standardized mixed date formats across all 99,457 records using Text to Columns (DMY format)
- Verified zero missing values and zero duplicate records across all columns
- Standardized all column headers for consistency
- Created Revenue column: Quantity × Price per transaction
- Created Age_Group column categorizing customers into 5 descriptive bands
- Exported cleaned dataset as CSV for SQL analysis and Power BI import

Age Group Classification

Age Group	Age Range	Definition
Youth	18–25	Early career / student shoppers
Young Adults	26–35	Career-building professionals
Middle Aged	36–45	Established professionals
Older Adults	46–55	Peak earning years
Seniors	56+	Retirement-age shoppers

Tools Used

- Microsoft Excel — Data cleaning, transformation, and feature engineering
- SQL Server (SSMS) Querying and aggregating 99,457 rows across 10 analytical questions
- Microsoft Power BI — Interactive 4-page dashboard with DAX measures and synced slicers

3. SQL Queries & Results

All analysis was performed using SQL Server Management Studio (SSMS) against the ShoppingDB database containing 99,457 rows in the customer_shopping table. Below are the 10 queries used to extract insights from the dataset.

Query 1 — Key Performance Indicators

```
SELECT
COUNT(*) AS Total_Transaction,
COUNT(DISTINCT Customer_ID) AS Total_Customers,
SUM(Quantity*Price) AS Total_Revenue,
ROUND(AVG(Quantity * Price), 2) AS Avg_Order_Value,
SUM(Quantity) AS Total_Item_Sold
FROM customer_shopping
```

Total Transactions	Total Revenue	Avg Order Value	Total Items Sold	Total Customers
99,457	\$251,505,794.25	\$2,528.79	298,712	99,457

Query 2 — Revenue by Product Category

```
SELECT
Category AS Product_Category,
COUNT(*) AS Total_Orders,
SUM(Quantity) AS Total_Items_sold,
ROUND(SUM(Quantity * Price), 2) AS Total_Revenue,
ROUND(AVG(Quantity * Price), 2) AS Avg_Order_Value
FROM customer_shopping
GROUP BY Category
ORDER BY Total_Revenue DESC
```

Category	Total Orders	Total Revenue
Clothing	34,487	\$113,996,791
Shoes	10,034	\$66,553,451
Technology	4,996	\$57,862,350
Cosmetics	9,946	\$6,792,863
Toys	7,210	\$3,980,426
Food & Beverage	22,792	\$849,535
Books	5,073	\$834,553
Souvenir	4,919	\$635,825

Query 3 — Monthly Revenue Trends

```
SELECT
YEAR(Invoice_Date) AS YEAR,
MONTH(Invoice_Date) AS Month_Number,
DATENAME(MONTH, Invoice_Date) AS Month_Name,
COUNT(*) AS Total_Orders,
ROUND(SUM(Quantity * Price), 2) AS Monthly_Revenue
FROM customer_shopping
GROUP BY
YEAR(Invoice_Date),
MONTH(Invoice_Date),
DATENAME(MONTH, Invoice_Date)
ORDER BY Year, Month_Number
```

Result: 2021 and 2022 each showed consistent monthly revenue of approximately \$9M-\$11M per month. January was consistently the strongest month. 2023 data runs only through March, with a sharp drop in transactions from April onward — confirmed as a dataset limitation, not a real business decline.

Query 4 — Revenue by Gender

```
SELECT
    Age_Group,
    COUNT(*) AS Total_Transactions,
    ROUND(SUM(quantity * price), 2) AS Total_Revenue,
    ROUND(AVG(quantity * price), 2) AS Avg_Order_Value
FROM customer_shopping
GROUP BY Age_Group
ORDER BY Total_Revenue DESC;
```

Gender	Total Customers	Total Revenue	Revenue %
Female	59,482	\$150,207,136	59.8%
Male	39,975	\$101,298,658	40.2%

Query 5 — Revenue by Age Group

```
SELECT
    Age_Group,
    COUNT(*) AS Total_Transactions,
    ROUND(SUM(quantity * price), 2) AS Total_Revenue,
    ROUND(AVG(quantity * price), 2) AS Avg_Order_Value
FROM customer_shopping
GROUP BY Age_Group
ORDER BY Total_Revenue DESC;
```

Age Group	Total Customers	Total Revenue
Seniors (56+)	20,169	\$67,103,885
Middle Aged (36-45)	19,952	\$50,184,236
Older Adults (46-55)	19,898	\$48,219,743
Young Adults (26-35)	19,745	\$47,879,660
Youth (18-25)	19,693	\$38,118,271

Query 6 — Payment Method Distribution

```
SELECT
    payment_method,
    COUNT(*) AS Total_Transactions,
    ROUND(SUM(quantity * price), 2) AS Total_Revenue,
    ROUND(COUNT(*) * 100.0 / SUM(COUNT(*)) OVER(), 2) AS Percentage_Share
FROM customer_shopping
GROUP BY payment_method
ORDER BY Total_Revenue DESC;
```

Payment Method	Total Revenue	Revenue %
Cash	\$112,832,243	44.69%
Credit Card	\$88,077,124	35.12%
Debit Card	\$50,596,427	20.19%

Query 7 — Revenue by Shopping Mall

```

SELECT
    shopping_mall,
    COUNT(*) AS Total_Transactions,
    COUNT(DISTINCT customer_id) AS Unique_Customers,
    ROUND(SUM(quantity * price), 2) AS Total_Revenue,
    ROUND(AVG(quantity * price), 2) AS Avg_Order_Value
FROM customer_shopping
GROUP BY shopping_mall
ORDER BY Total_Revenue DESC;

```

Shopping Mall	Total Transactions	Total Revenue
Mall of Istanbul	19,943	\$50,872,482
Kanyon	19,823	\$50,554,231
Metrocity	15,011	\$37,302,787
Metropol AVM	19,161	\$25,379,913
Istinye Park	9,781	\$24,618,828
Zorlu Center	5,075	\$12,901,054
Cevahir AVM	4991	\$12,645,138
Viaport Outlet	4918	\$12,521,340
Emaar Square Mall	4811	\$12,406,100
Forum Istanbul	4949	\$12,303,921

Query 8 — Top Category Per Mall

```

SELECT
    shopping_mall,
    category,
    ROUND(SUM(quantity * price), 2) AS Revenue
FROM customer_shopping
GROUP BY shopping_mall, category
ORDER BY shopping_mall, Revenue DESC;

```

💡 Query 8 Result

Clothing ranked as the #1 revenue category in every single one of the 10 malls without exception — confirming its universal dominance across Istanbul's entire mall ecosystem.

Query 9 — Year over Year Revenue

```
SELECT
    YEAR(invoice_date) AS Year,
    COUNT(*) AS Total_Orders,
    ROUND(SUM(quantity * price), 2) AS Total_Revenue,
    ROUND(AVG(quantity * price), 2) AS Avg_Order_Value
FROM customer_shopping
GROUP BY YEAR(invoice_date)
ORDER BY Year;
```

Year	Total Order	Total Revenue	Avg Order Value
2021	45,382	\$114,560,571	45,382
2022	45,551	\$115,436,814	45,551
2023	8524	\$21,508,410	8524

Query 10 — Top 10 Customers by Revenue

```
SELECT TOP 10
    customer_id,
    COUNT(*) AS Total_Purchases,
    ROUND(SUM(quantity * price), 2) AS Lifetime_Value,
    ROUND(AVG(quantity * price), 2) AS Avg_Order_Value
FROM customer_shopping
GROUP BY customer_id
ORDER BY Lifetime_Value DESC;

USE ShoppingDB;
SELECT * FROM customer_shopping;
```

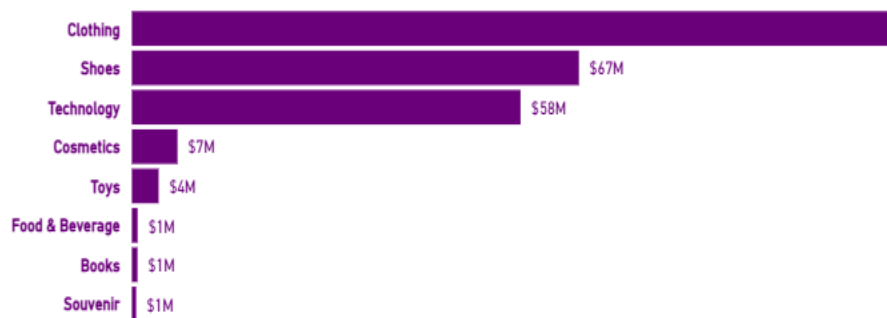
💡 Query 10 Result — Anomaly Detected

All top 10 customers spent exactly \$26,250 each. This uniform spending pattern across unrelated customers is statistically unusual and likely indicates bulk/wholesale purchasing or a data generation artifact. This was flagged as a data quality observation in the final report.

4. Key Findings & Analysis

4.1 Revenue by Product Category

Clothing is the dominant revenue category, generating \$113,996,791 — nearly double the second-ranked category, Shoes at \$66,553,451. Technology ranks third at \$57,862,350 but records the highest average order value at \$11,581, suggesting high-value individual purchases.



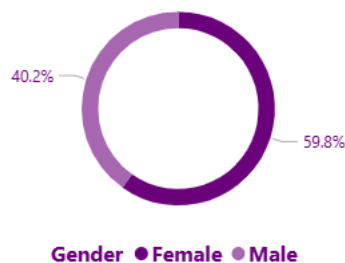
Rank	Category	Total Revenue	Avg Order Value
1	Clothing	\$113,996,791	\$3,308
2	Shoes	\$66,553,451	\$6,633
3	Technology	\$57,862,350	\$11,581
4	Cosmetics	\$6,792,863	\$683
5	Toys	\$3,980,426	\$552

💡 Strategic Insight — Q1 Answer

Clothing should remain the anchor category in every mall's tenant mix. However, Technology's exceptionally high average order value makes it a premium revenue driver worth investing in — dedicated tech zones or expanded tech retailer space could significantly boost per-visit revenue.

4.2 Gender Analysis

Female customers account for 59.8% of total revenue (\$150,207,136) compared to males at 40.2% (\$101,298,658). Female shoppers also make up the majority of transactions at 59,482 versus 39,975 male transactions, indicating higher shopping frequency rather than higher spend per visit.



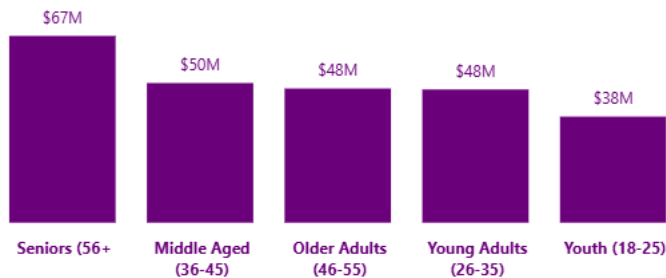
Gender	Total Revenue	Revenue Share	Total Transactions
Female	\$150,207,136	59.8%	59,482
Male	\$101,298,658	40.2%	39,975

 **Strategic Insight — Q2 Answer**

Female shoppers are the primary revenue drivers. Marketing campaigns, loyalty programs, and promotional events should be designed with female shopping preferences as the primary target. However, male shoppers represent a significant 40% share — a largely untapped growth segment with potential for targeted campaigns around Technology and Sports categories.

4.3 Age Group Analysis

Seniors aged 56 and above are the highest revenue-generating age group at \$67,103,885, despite being a typically underestimated demographic. Middle Aged customers (36–45) follow at \$50,184,236, while Youth (18–25) record the lowest revenue at \$38,118,271.



Rank	Age Group	Total Revenue
1	Seniors (56+)	\$67,103,885
2	Middle Aged (36–45)	\$50,184,236
3	Older Adults (46–55)	\$48,219,743

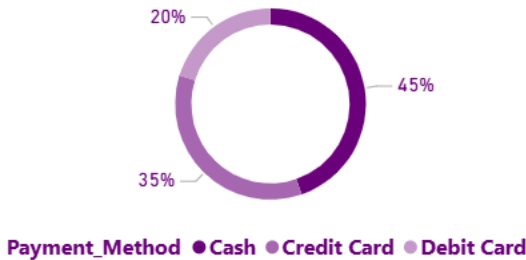
4	Young Adults (26–35)	\$47,879,660
5	Youth (18–25)	\$38,118,271

 Strategic Insight — Q3 Answer

Seniors are the highest-value customer segment and are frequently overlooked in retail marketing. Malls should consider senior-friendly initiatives — comfortable seating, accessible store layouts, senior loyalty discounts, and weekday-targeted promotions when malls are less crowded. This could deepen engagement with the most valuable demographic.

4.4 Payment Method Preferences

Cash remains the most preferred payment method at 44.69% of total revenue (\$112,832,243), followed by Credit Card at 35.12% (\$88,077,124) and Debit Card at 20.19% (\$50,596,427). The continued dominance of cash is notable in the context of global digital payment trends.



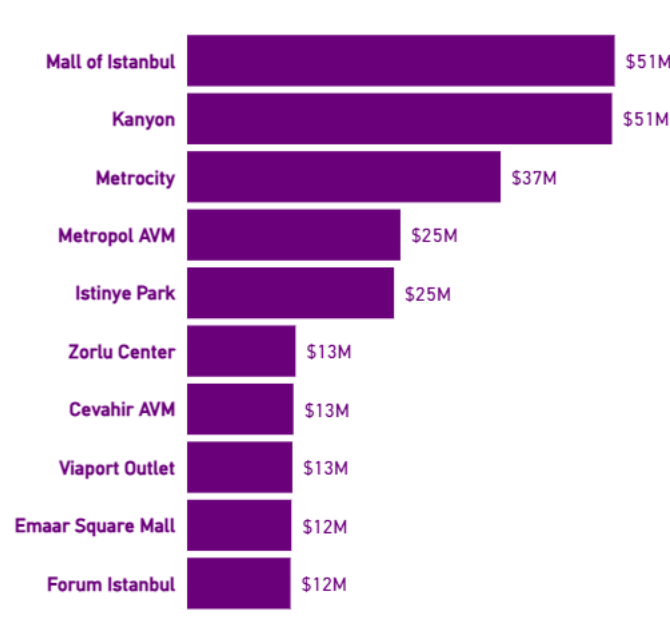
Payment Method	Total Revenue	Revenue Share
Cash	\$112,832,243	44.69%
Credit Card	\$88,077,124	35.12%
Debit Card	\$50,596,427	20.19%

 Strategic Insight — Q4 Answer

The high cash usage (44.69%) may reflect a segment of older shoppers or tourists uncomfortable with digital payments. Mall management should ensure sufficient ATM availability and cash-handling infrastructure. Simultaneously, incentivizing card and digital payments through loyalty point bonuses could help modernize payment behavior and reduce cash handling costs.

4.5 Shopping Mall Performance

Mall of Istanbul leads all malls in revenue at \$50,872,482, closely followed by Kanyon at \$50,554,231. Together, these two malls account for approximately 40% of total revenue. Metrocity ranks third at \$37,302,787, while smaller malls like Forum Istanbul and Emaar Square Mall generate approximately \$12M each.



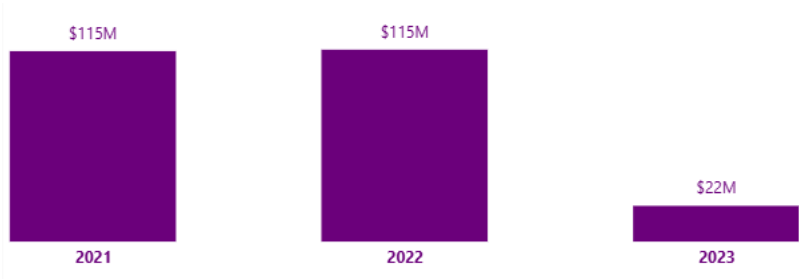
Rank	Shopping mall	Total Revenue
1	Mall of Istanbul	\$50,872,482
2	Kanyon	\$50,554,231
3	Metrocity	\$37,302,787
4	Metropol AVM	\$25,000,000+
5	Istinye Park	\$25,000,000+

💡 Strategic Insight — Q5 Answer

Clothing is the #1 revenue category in every single mall without exception — confirming that fashion retail is the universal anchor across Istanbul's mall ecosystem. Mall of Istanbul and Kanyon's near-identical revenue figures suggest direct competition for the same customer base. Differentiating through exclusive tenant offerings, experiential retail, or niche category expansion could be key to competitive separation.

4.6 Revenue Trends Over Time

Annual revenue was broadly consistent across the full data period. 2021 generated \$114,560,571 and 2022 recorded the highest full-year revenue at \$115,436,814 — a marginal 0.76% year-on-year increase. Monthly revenue data shows January as the strongest month (\$25.8M) with a gradual decline through mid-year before recovering in October (\$21.0M).



Year	Total Revenue	YoY Change
2021	\$114,560,571	Baseline
2022	\$115,436,814	+0.76%
2023	\$21,508,410	Partial data (Jan–Mar only)

 **Important Data Note — Q6 Answer**

The sharp revenue drop in 2023 is NOT a genuine business decline. The dataset contains complete data only through March 2023, after which transaction records stop abruptly. This is a dataset limitation, not a real trend. Any business analysis comparing 2023 to prior years must account for this incomplete data period.

4.7 Top Customer Anomaly

Analysis of the top 10 customers by revenue revealed that all 10 spent exactly \$26,250 each. This level of uniformity is statistically unusual and may indicate bulk purchasing, wholesale accounts, or a data generation artifact in the original dataset.

 **Strategic Insight — Q7 Answer**

Mall management should investigate whether these top-spending accounts represent genuine retail customers or wholesale/trade buyers. If wholesale activity is occurring within retail mall space, this may require separate account management, pricing structures, or dedicated B2B services.

5. Strategic Recommendations

Based on the analysis findings, the following recommendations are proposed for Istanbul shopping mall management:

1. Double Down on Clothing — But Diversify Strategically

Clothing dominates revenue universally. However, Technology's high average order value (\$11,581) presents a significant upsell opportunity. Consider expanding premium tech retail space in high-footfall malls to capture higher per-visit spending.

2. Build a Senior Loyalty Programme

Seniors (56+) are the highest revenue segment yet are frequently overlooked in mall marketing. A dedicated senior loyalty programme with weekday discounts, accessible facilities, and targeted communications could deepen engagement with the most valuable customer group.

3. Target Male Shoppers as a Growth Segment

Male shoppers represent 40% of revenue — a substantial segment with room to grow. Targeted campaigns focused on Technology, Sports, and premium categories preferred by male shoppers could close the revenue gap and diversify the customer base.

4. Modernize Payment Infrastructure

Cash dominates at 44.69% higher than expected for an urban mall environment. Introducing digital payment incentives (loyalty points for card use, contactless payment promotions) could reduce cash handling costs while improving transaction speed and customer experience.

5. Investigate Top Customer Accounts

The uniformity of top 10 customer spending (\$26,250 each) warrants further investigation. Understanding whether these are genuine retail customers or wholesale buyers will help management tailor account management strategy appropriately.

6. Seasonal Campaign Planning

January is consistently the strongest revenue month (\$25.8M). Management should plan premium launch campaigns, exclusive offers, and high-profile tenant activations in January to maximize the natural revenue peak. Mid-year dip months (July–August) should be addressed with targeted promotions and events to sustain footfall.

6. Conclusion

This analysis of 99,457 transactions across 10 Istanbul shopping malls reveals a retail ecosystem with consistent performance, clear category leadership, and identifiable customer segments with distinct spending patterns.

The dominance of Clothing, the spending power of Senior customers, the revenue contribution of female shoppers, and the persistent preference for cash payments are four defining characteristics of Istanbul's mall retail landscape. Each of these findings carries actionable implications for mall management strategy.

The 2023 data incompleteness should be noted in all future reporting, and the top customer spending anomaly should be investigated to ensure accurate customer segmentation.

With targeted action on tenant mix, customer loyalty, payment modernization, and seasonal planning, Istanbul's shopping malls are well-positioned to sustain and grow revenue performance in the years ahead.

Prepared by Ijeoma Okeke | Data Analyst

July 2026 | Istanbul Shopping Analytics Portfolio Project